You may be asking a more basic question: what exactly is AWS Lambda? Before diving into our first real-world example, let us introduce you, briefly, to Lambda and explain why it's often mentioned in the context of an architecture that's being called *serverless*.

6.1 Executing your code with AWS Lambda

Computing capacity is available at different layers of abstraction on AWS: virtual machines, containers, and functions. You learned about the virtual machines offered by Amazon's EC2 service in chapter 3. Containers offer another layer of abstraction on top of virtual machines; you will learn about containers in chapter 18. AWS Lambda provides computing power, as well, but in a fine-grained manner: it is an execution environment for small functions, rather than a full-blown operating system or container.

6.1.1 What is serverless?

When reading about AWS Lambda, you might have stumbled upon the term *serverless*. In his book *Serverless Architectures on AWS* (Manning, 2022; http://mng.bz/wyr5), Peter Sbarski summarizes the confusion created by this catchy and provocative phrase:

> *[...] the word serverless is a bit of a misnomer. Whether you use a compute service such as AWS Lambda to execute your code, or interact with an API, there are still servers running in the background. The difference is that these servers are hidden from you. There's no infrastructure for you to think about and no way to tweak the underlying operating system. Someone else takes care of the nitty-gritty details of infrastructure management, freeing your time for other things.*